



MODUL **PRAKTIKUM**

ADMINISTRASI BASIS DATA

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Modul Praktikum 13

Monitoring Performa SQL Server Menggunakan Activity Monitor, PerfMon, dan SQL Server Profiler

A. TUJUAN

Setelah mengikuti praktikum ini mahasiswa mampu:

- Menggunakan Activity Monitor untuk memantau aktivitas SQL Server.
- Mengidentifikasi query yang menggunakan resource dalam jumlah besar.
- Menggunakan Performance Monitor (PerfMon) untuk memantau CPU, memori, dan disk.
- Menggunakan SQL Server Profiler untuk merekam aktivitas kueri.
- Mengidentifikasi kueri yang lambat (slow query).
- Memahami konsep blocking dan deadlock.
- Melakukan analisis dasar performa database.

B. ALAT DAN BAHAN

Software:

- Microsoft SQL Server
- SQL Server Management Studio (SSMS)
- SQL Server Profiler
- Windows Performance Monitor (PerfMon)

Database

Gunakan database praktikum.

C. STUDI KASUS

Sistem Akademik Universitas memiliki database yang digunakan oleh banyak pengguna secara bersamaan.

Administrator database ingin:

- Mengetahui aktivitas SQL Server secara real-time.
- Mengetahui query yang paling banyak menggunakan CPU.
- Mengetahui query yang berjalan lambat.
- Mengidentifikasi blocking dan deadlock.
- Memantau penggunaan resource server.

D. PRAKTIKUM

PRAKTIKUM 1

Monitoring SQL Server Menggunakan Activity Monitor

Langkah 1

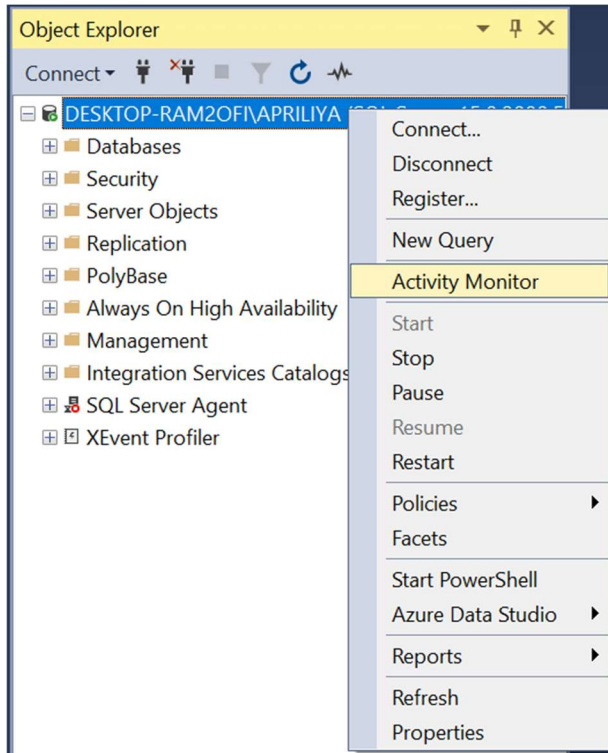
Buka SQL Server Management Studio.

Pilih:

Object Explorer → Klik kanan nama server → Activity Monitor

Atau:

Ctrl + Alt + A



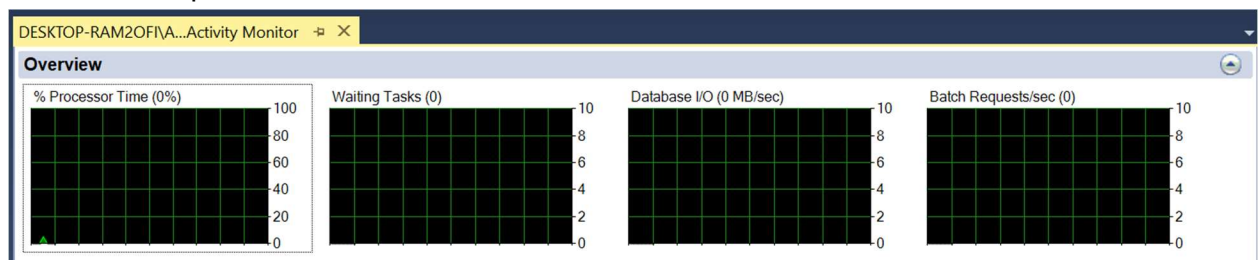
Langkah 2

Amati bagian:

Overview

Perhatikan informasi:

- CPU Usage
- Waiting Tasks
- Database I/O
- Batch Requests/sec



Langkah 3

Jalankan query berikut:

```
SELECT *  
FROM Akademik.Mahasiswa;
```

Langkah 4

Amati perubahan aktivitas pada Activity Monitor.

Hasil Pengamatan

Parameter	Nilai
CPU Usage	5%
Waiting Tasks	0
Database I/O	0 MB/sec
Batch Requests/sec	0

PRAKTIKUM 2

Melihat Query yang Menggunakan Resource Tinggi

Langkah 1

Buka:

Activity Monitor → Recent Expensive Queries

Recent Expensive Queries							
Query	Exe...	CP...	Phy...	Logi...	Logi...	Ave...	Pla...
WITH merged_query_stats AS...	4	1	9	0	19	19	1 temp...

Langkah 2

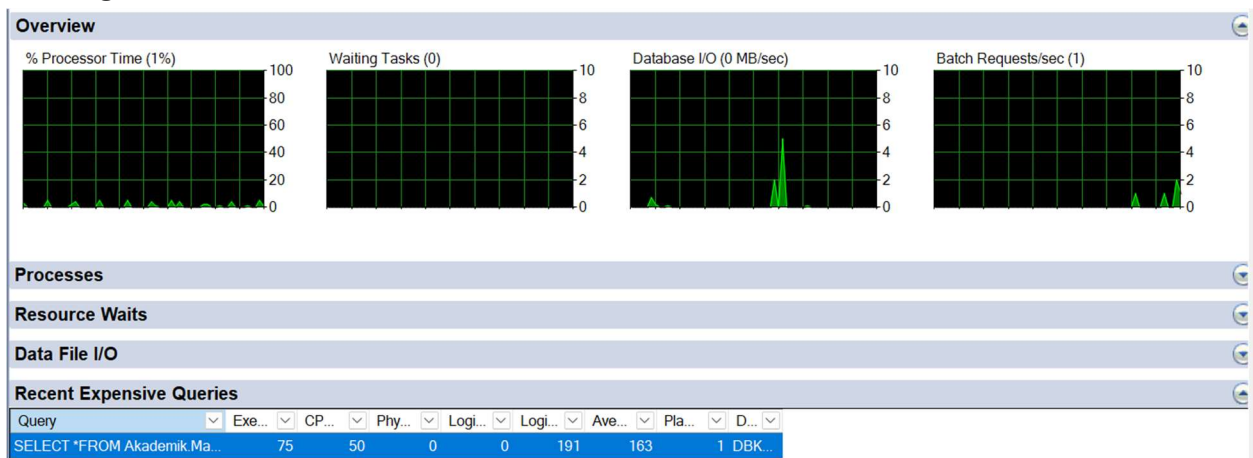
Jalankan query berikut beberapa kali:

```
SELECT *  
FROM Akademik.Mahasiswa  
ORDER BY Nama;
```

Langkah 3

Perhatikan:

- CPU
- Duration
- Logical Reads



Hasil Pengamatan

Query	CPU	Duration
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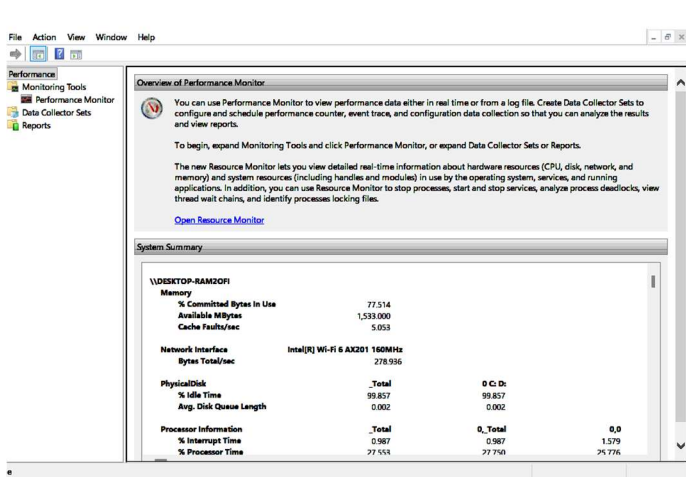
	1%	
--	----	--

PRAKTIKUM 3

Monitoring CPU dan Memory Menggunakan PerfMon

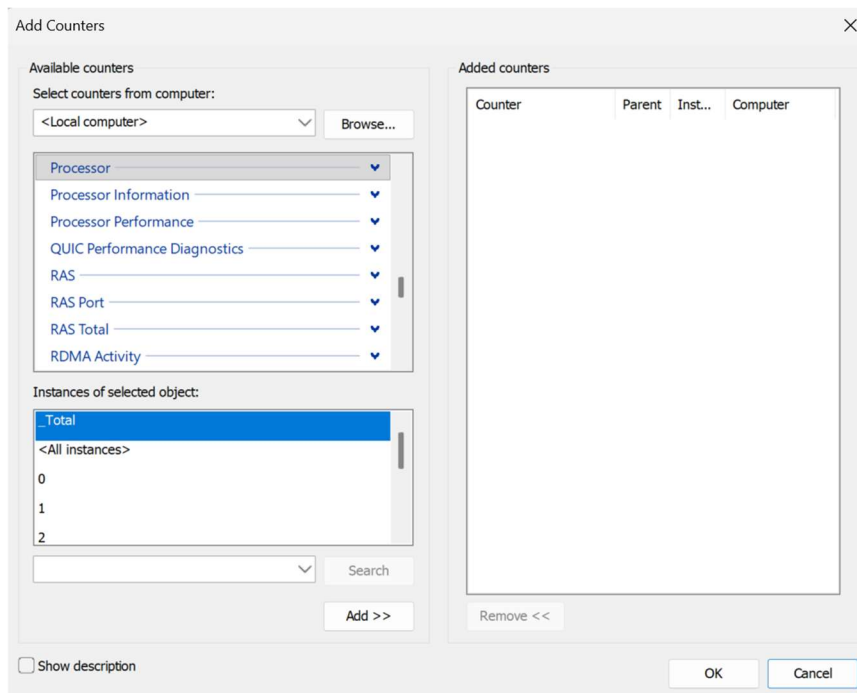
Langkah 1

- Tekan: Windows + R
- Ketik: Perfmon



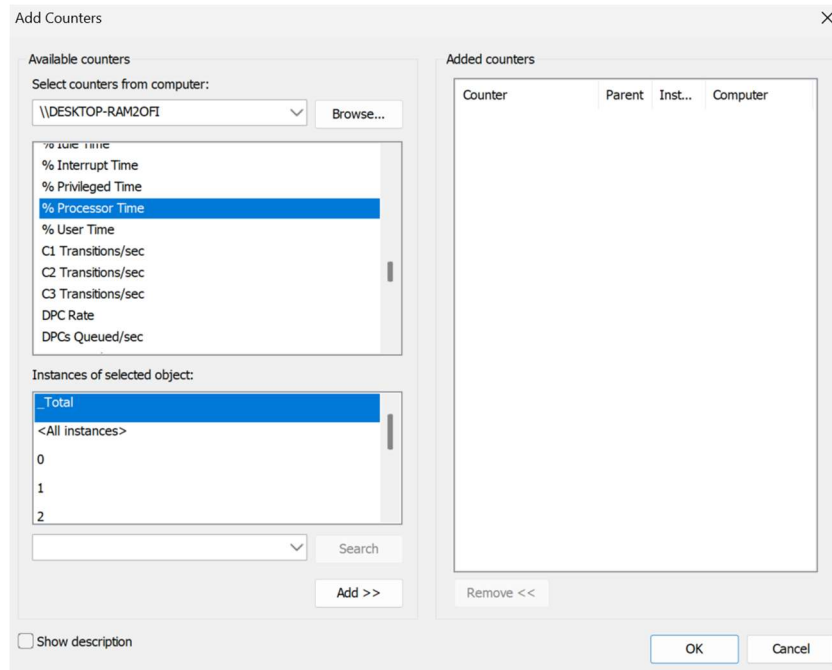
Langkah 2

- Masuk ke: Performance Monitor
- Klik: +

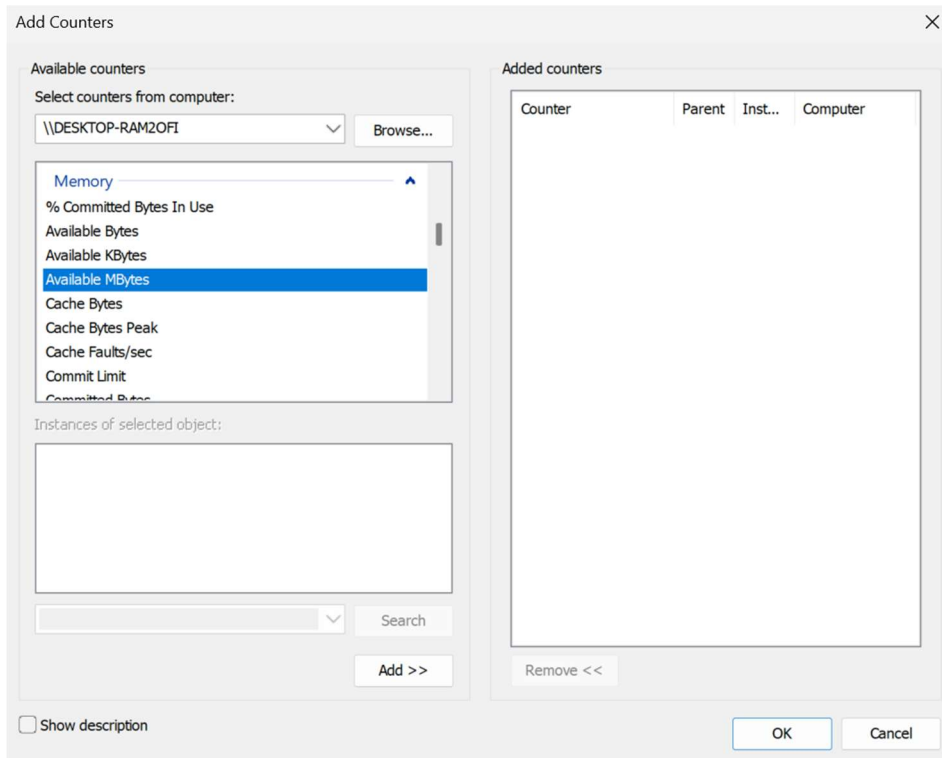


Langkah 3

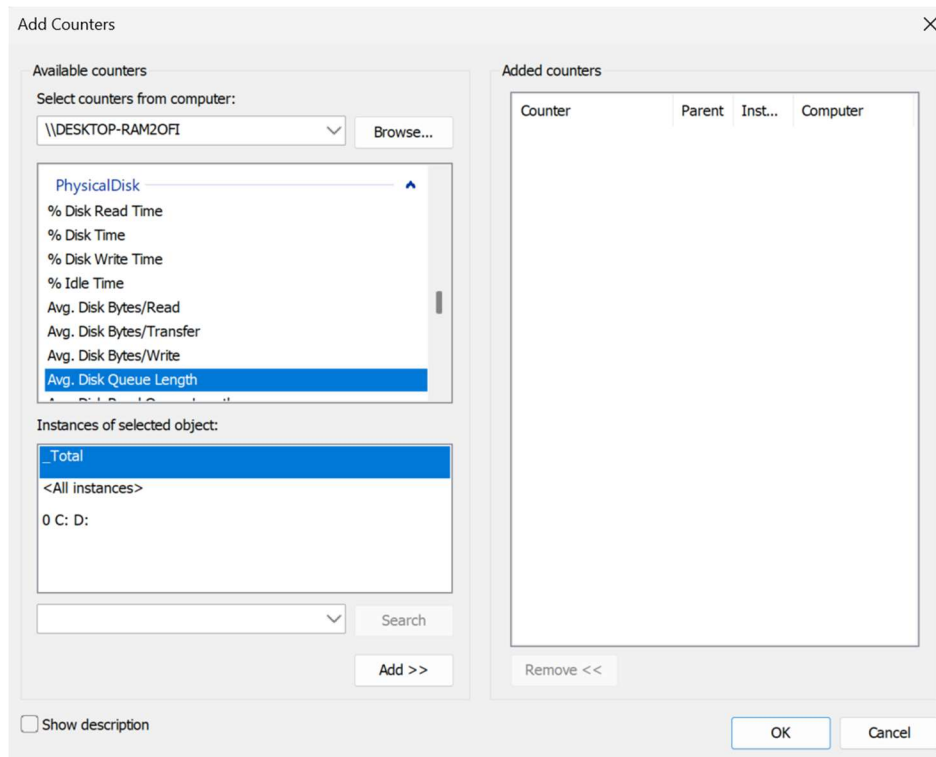
- Tambahkan Counter: CPU → Processor(_Total)\% Processor Time



- Memory: Memory\\Available Mbytes



- Disk: PhysicalDisk\\Avg. Disk Queue Length

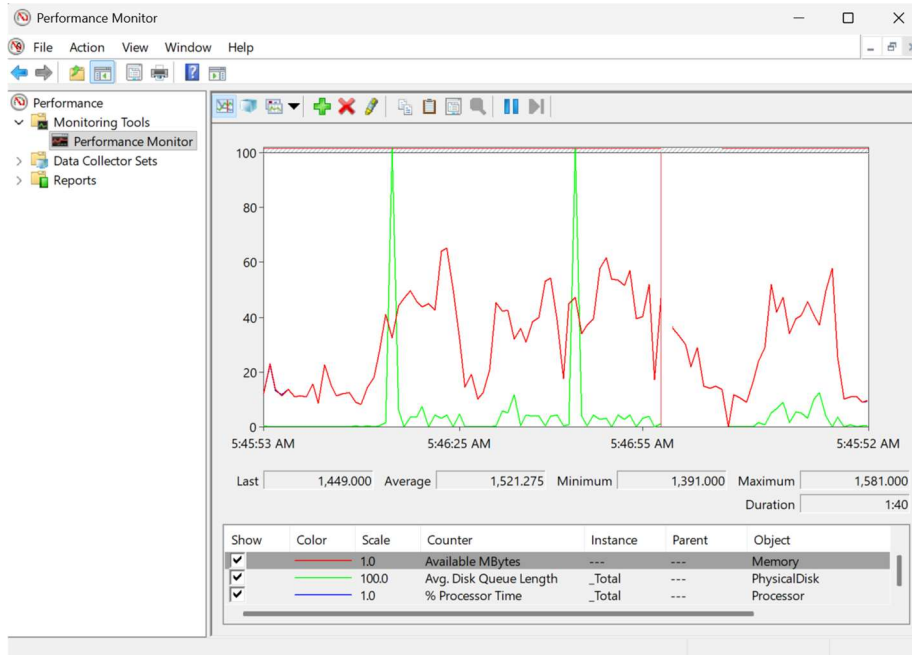
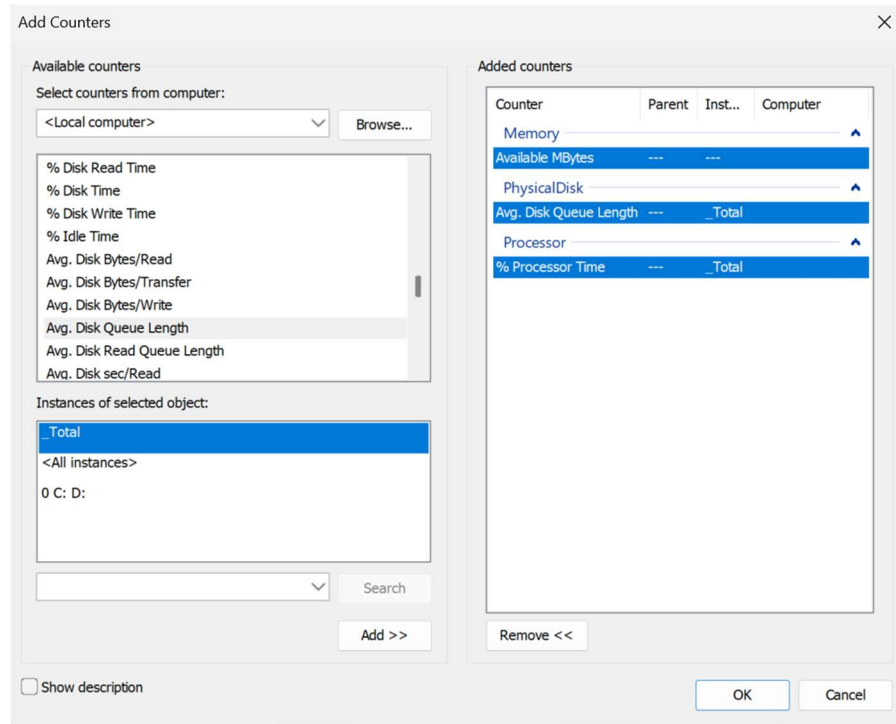


Langkah 4

Jalankan query berikut berulang kali:

```
SELECT *  
FROM Akademik.Mahasiswa;  
GO 50
```

Hasil Pengamatan

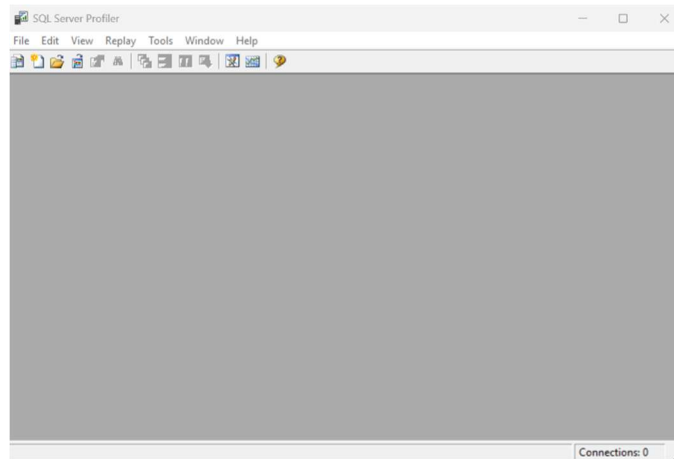


PRAKTIKUM 4

Menggunakan SQL Server Profiler

Langkah 1

Buka: Tools → SQL Server Profiler



Langkah 2

Pilih: File → New Trace

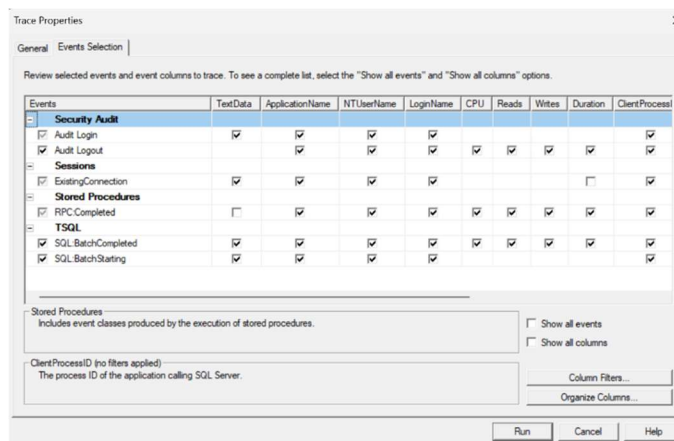
Langkah 3

Gunakan Template: Standard

Langkah 4

Centang event:

- SQL: BatchCompleted
- RPC: Completed



Langkah 5

Klik: Run

Langkah 6

Kembali ke SSMS.

Jalankan:

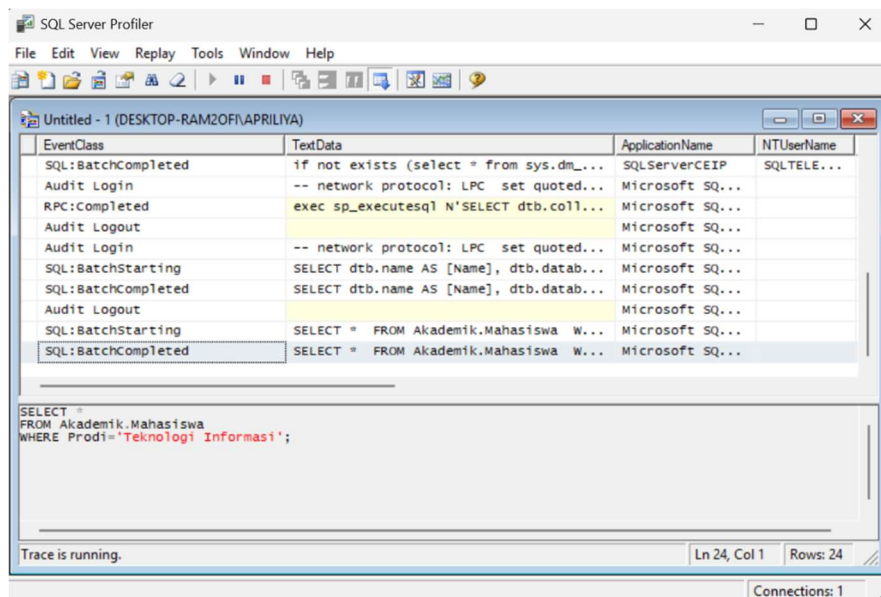
```
SELECT *
FROM Akademik.Mahasiswa
WHERE Prodi='Teknologi Informasi';
```

Hasil

Amati:

- Query yang dieksekusi

- Start Time
- End Time
- Duration



PRAKTIKUM 5

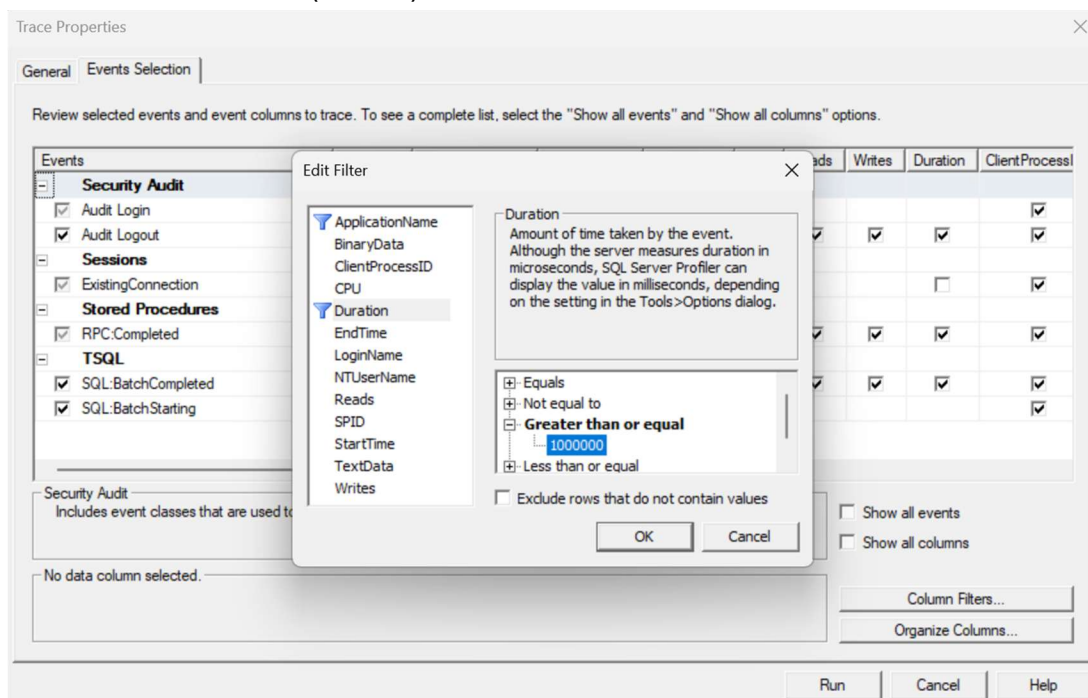
Mencari Query yang Lambat

Langkah 1

Pada Profiler pilih: Column Filters

Langkah 2

Filter: Duration > 1000000 (1 detik)



Langkah 3

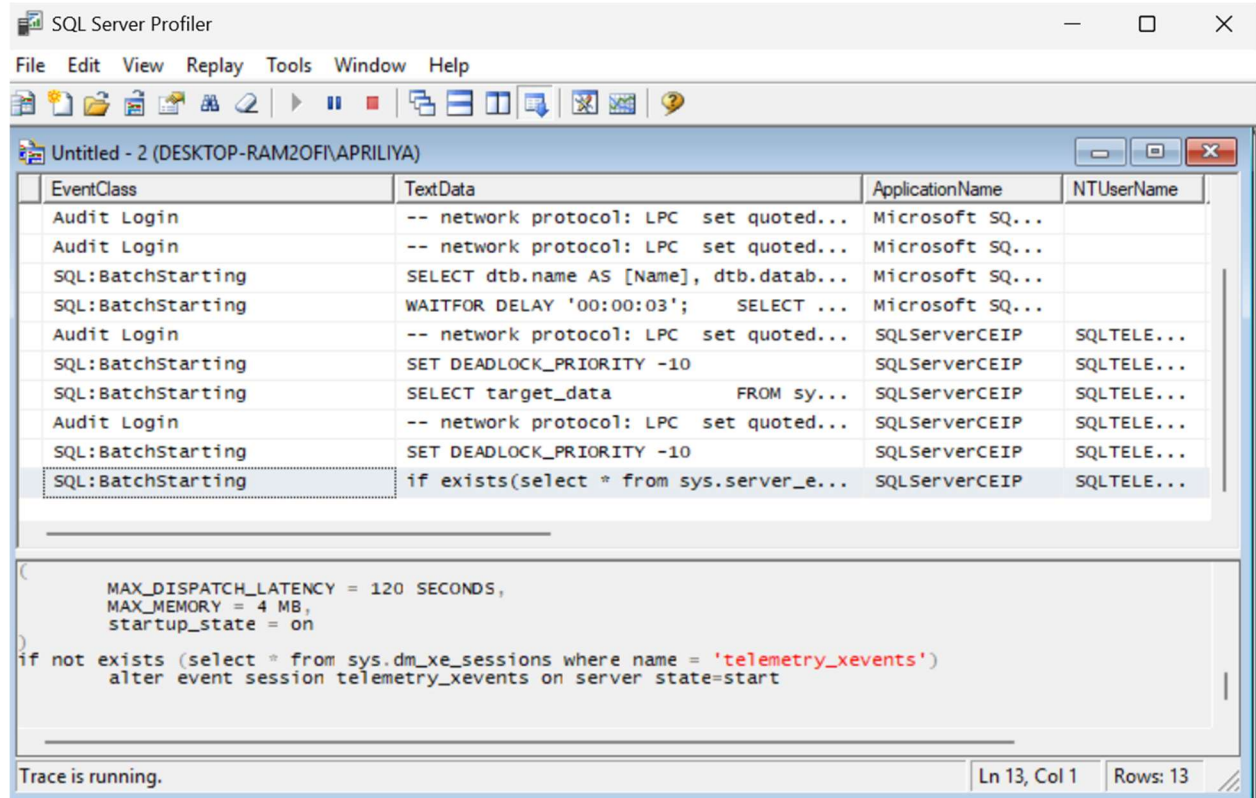
Jalankan query:

```
WAITFOR DELAY '00:00:03';
```

```
SELECT *  
FROM Akademik.Mahasiswa;
```

Langkah 4

Amati hasil pada Profiler.



SQL Server Profiler window showing a trace of SQL queries. The trace is titled "Untitled - 2 (DESKTOP-RAM2OF\APRILIYA)". The trace is running, and the status bar indicates "Trace is running." The trace shows the following events:

EventClass	TextData	ApplicationName	NTUserName
Audit Login	-- network protocol: LPC set quoted...	Microsoft SQ...	
Audit Login	-- network protocol: LPC set quoted...	Microsoft SQ...	
SQL:BatchStarting	SELECT dtb.name AS [Name], dtb.datab...	Microsoft SQ...	
SQL:BatchStarting	WAITFOR DELAY '00:00:03'; SELECT ...	Microsoft SQ...	
Audit Login	-- network protocol: LPC set quoted...	SQLServerCEIP	SQLTELE...
SQL:BatchStarting	SET DEADLOCK_PRIORITY -10	SQLServerCEIP	SQLTELE...
SQL:BatchStarting	SELECT target_data FROM sy...	SQLServerCEIP	SQLTELE...
Audit Login	-- network protocol: LPC set quoted...	SQLServerCEIP	SQLTELE...
SQL:BatchStarting	SET DEADLOCK_PRIORITY -10	SQLServerCEIP	SQLTELE...
SQL:BatchStarting	if exists(select * from sys.server_e...	SQLServerCEIP	SQLTELE...

The selected batch (SQL:BatchStarting) shows the following SQL query:

```
MAX_DISPATCH_LATENCY = 120 SECONDS,  
MAX_MEMORY = 4 MB,  
startup_state = on  
)  
if not exists (select * from sys.dm_xe_sessions where name = 'telemetry_xevents')  
alter event session telemetry_xevents on server state=start
```

Trace is running. Ln 13, Col 1 Rows: 13